Maximum and minimum values

1

a $x = 1, 3$

b $(1, 11)$

c $(3, 3)$

d 227

e 3

2

a $(0, 4)$

b $\left(-\frac{3}{5}, \frac{91}{25}\right), \left(-\frac{1}{3}, \frac{97}{27}\right)$

c $\left(-\frac{3}{5}, \frac{91}{25}\right)$ is a local maximum and $\left(-\frac{1}{3}, \frac{97}{27}\right)$ is a local minimum.

d As $f''(x) = 0$ and the concavity changes from negative to positive, there is a point of inflection at $x = -\frac{7}{15}$. The point of inflection is $\left(-\frac{7}{15}, \frac{2441}{675}\right)$.

e As $x \to \infty, y \to \infty$ and as $x \to -\infty, y \to -\infty$.

f The minimum is 3 and the maximum is 19.

3

a $(0, 5)$

b None

c Even

d $(-1, 4), (0, 5), (1, 4)$

e $(0, 5)$ is a local maximum, and $(1, 4)$ and $(-1, 4)$ are global minimums.

f As $f''(x) = 0$ and the concavity changes from positive to negative at $x = -\frac{\sqrt{3}}{3}$ and from negative to positive at $x = \frac{\sqrt{3}}{3}$. The points of inflection are $\left(-\frac{\sqrt{3}}{3}, \frac{40}{9}\right)$ and $\left(\frac{\sqrt{3}}{3}, \frac{40}{9}\right)$.

g As $x \to \infty, y \to \infty$ and as $x \to -\infty, y \to \infty$.

h The maximum is 13 and the minimum is 4.

4

a $(0, -1)$

b $(-2, -33), (0, -1), (1, -6)$

c $(0, -1)$ is a local maximum, $(1, -6)$ is a local minimum and $(-2, -33)$ is a global minimum.

d As $f''(x) = 0$ and the concavity changes from positive to negative at $x = -1.22$ and from negative to positive at $x = 0.55$. The points of inflection are $(-1.22, -19.36)$ and $(0.55, -3.68)$.

e As $x \to \infty, y \to \infty$ and as $x \to -\infty, y \to \infty$.

f The maximum is -1 and the minimum is -14.

Graphs of functions

5

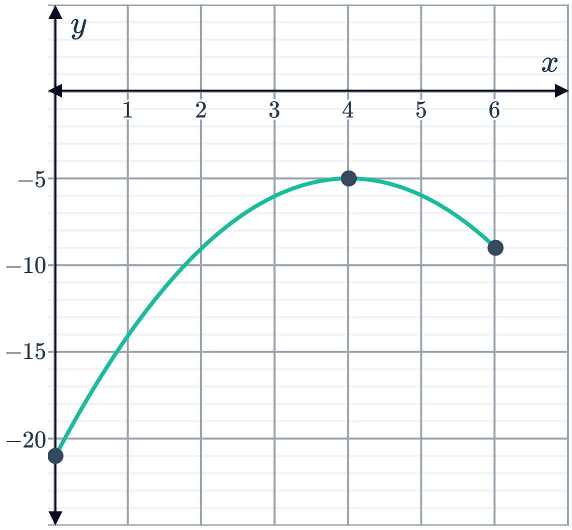
a $(0, -21)$

b $(4, -5)$

c $(4, -5)$ is a global maximum point.

d The maximum value is -5 and the minimum value is -21.

e



6

a (0,1)

b $(-1,0),(1,0)$

c Even

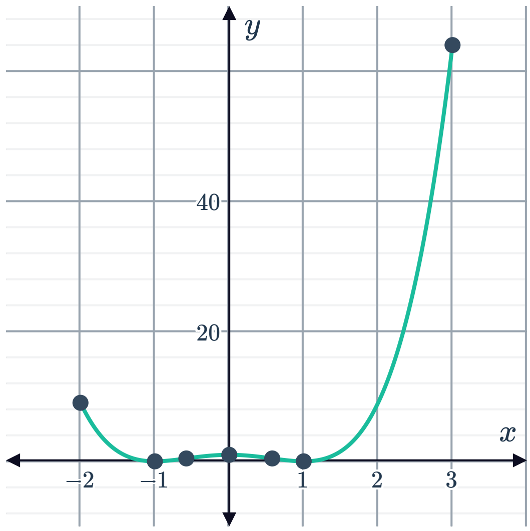
d $(-1,0),(0,1),(1,0)$

e $(-1,0)$ and $(1,0)$ are global minimum and $(0,1)$ is a local maximum.

f As $f''(x)=0$ and the concavity changes from positive to negative at $x=-\frac{\sqrt{3}}{3}$ and negative to positive at $x=\frac{\sqrt{3}}{3}$. The points of inflection are $\left(-\frac{\sqrt{3}}{3},\frac{4}{9}\right)$ and $\left(\frac{\sqrt{3}}{3},\frac{4}{9}\right)$.

g The maximum is 64 and the minimum is 0.

h



7

a $(0,1)$

b $\left(\frac{1}{2},0\right),(1,0)$

c $\left(\frac{1}{2},0\right),\left(\frac{5}{6},\frac{2}{27}\right)$

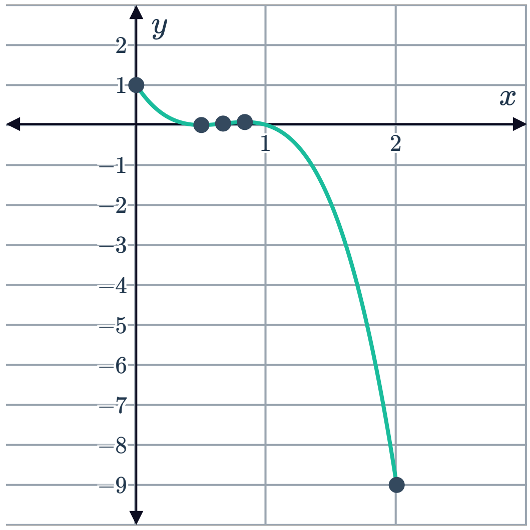
d $\left(\frac{1}{2},0\right)$ is a local minimum and $\left(\frac{5}{6},\frac{2}{27}\right)$ is a local maximum.

e As $f''(x)=0$ and the concavity changes from positive to negative, there is a point of inflection at $x=\frac{2}{3}$. The point of inflection is $\left(\frac{2}{3},\frac{1}{27}\right)$.

f As $x\to\infty, y\to-\infty$ and as $x\to-\infty, y\to\infty$.

g The maximum is 1 and the minimum is -9.

h



8

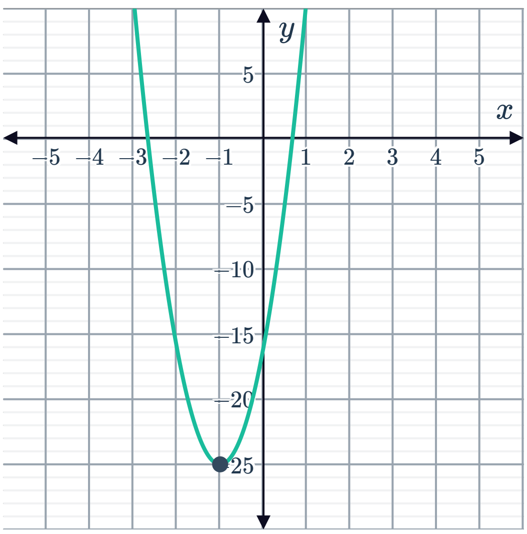
a $(0,-16)$

b $\left(-\frac{8}{3},0\right),\left(\frac{2}{3},0\right)$

c $(-1,-25)$

d $(-1,-25)$ is the global minimum point.

e



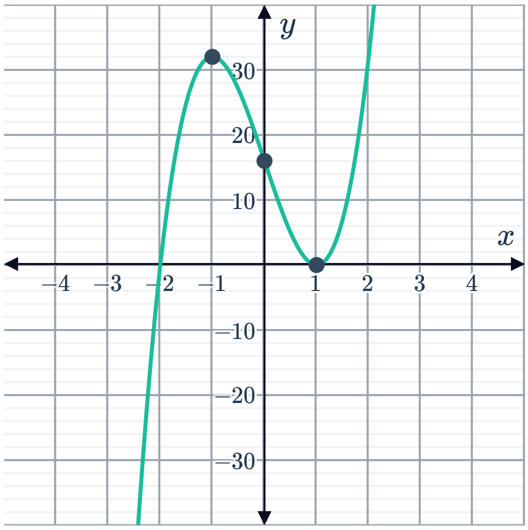
9

a $(-1, 32), (1, 0)$

b $(-1, 32)$ is a local maximum and $(1, 0)$ is a local minimum.

c $(0, 16)$

d



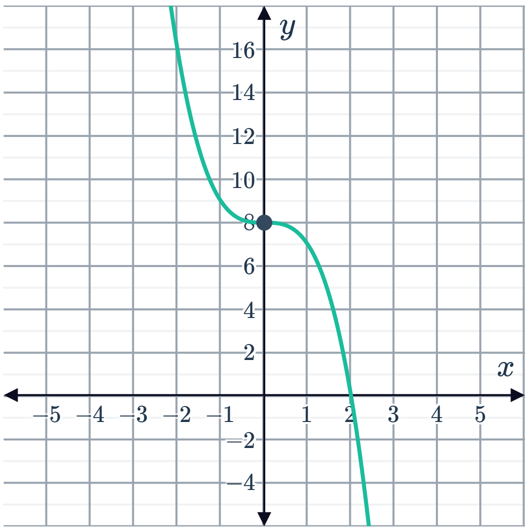
10

a

i $(0, 8)$ is a horizontal point of inflection.

ii $(0, 8)$

iii



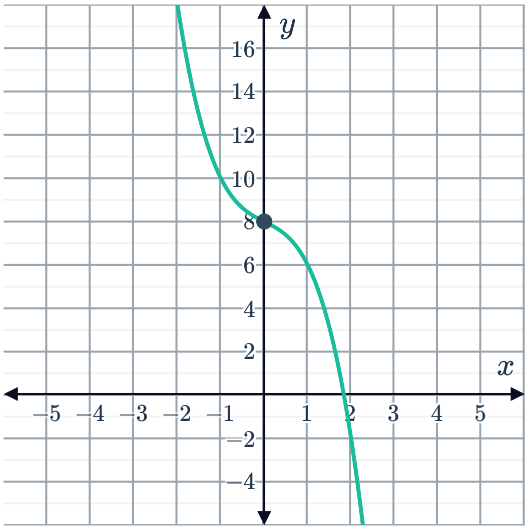
iv $(-\infty, \infty)$

b

i None

ii $(0, 8)$

iii



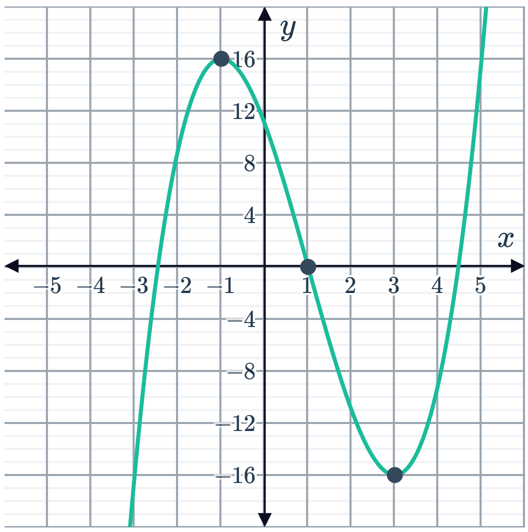
iv $(-\infty, \infty)$

c

i $(-1, 16)$ is a local maximum and $(3, -16)$ is a local minimum.

ii $(1, 0)$

iii



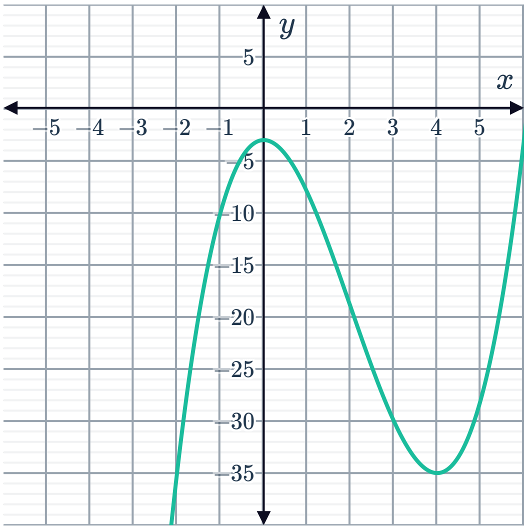
iv $(-1, 3)$

d

i $(0,-3)$ is a local maximum and $(4,-35)$ is a local minimum.

ii $(2,-19)$

iii



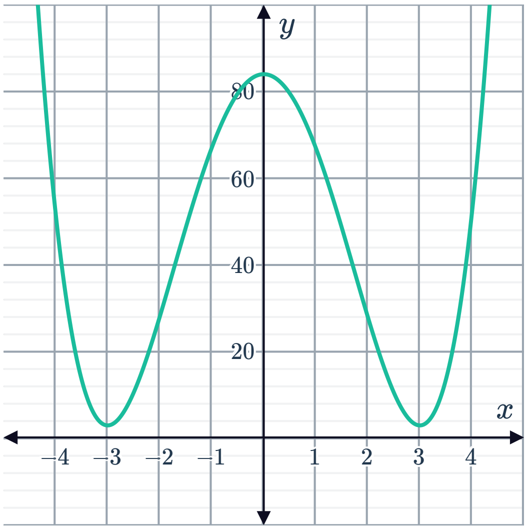
iv $(0,4)$

e

i $(-3,3)$ and $(3,3)$ are global minimums, and $(0,84)$ is a local maximum

ii $\left(-\sqrt{3},39\right),\left(\sqrt{3},39\right)$

iii



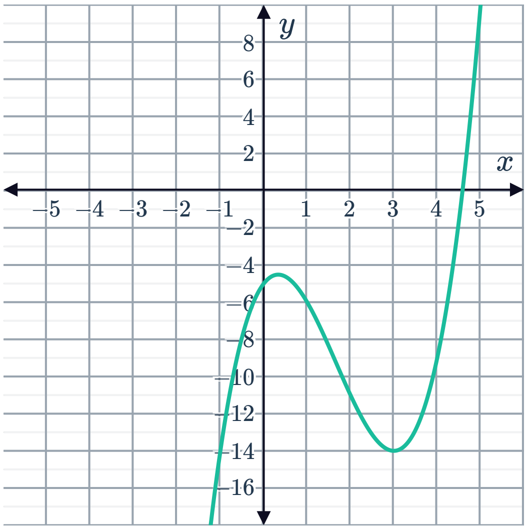
iv $(-\infty,-3)\cup(0,3)$

f

i $\left(\frac{1}{3}, -\frac{122}{27}\right)$ is a local maximum and $(3, -14)$ is a local minimum.

ii $\left(\frac{5}{3}, -\frac{250}{27}\right)$

iii



iv $\left(\frac{1}{3}, 3\right)$

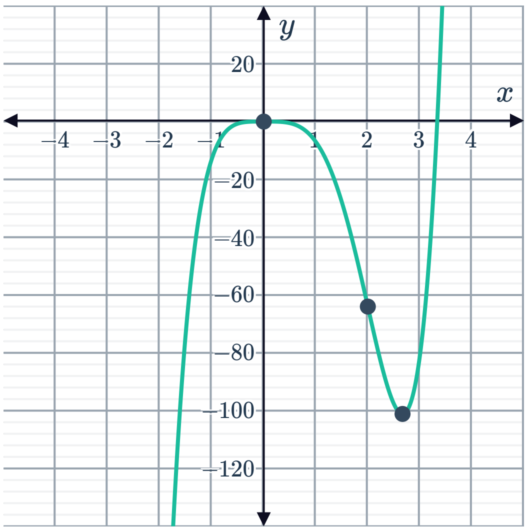
11

a

i $(0, 0)$ is a local maximum and $\left(\frac{8}{3}, -\frac{8192}{81}\right)$ is a local minimum.

ii $(2, -64)$

iii



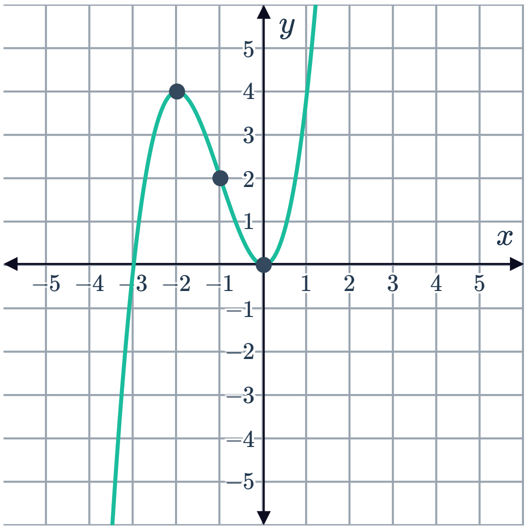
iv $(-\infty, 0) \cup \left(\frac{8}{3}, \infty\right)$

b

i $(-2,4)$ is a local maximum and $(0,0)$ is a local minimum.

ii $(-1,2)$

iii



iv $(-\infty,-2) \cup (0,\infty)$

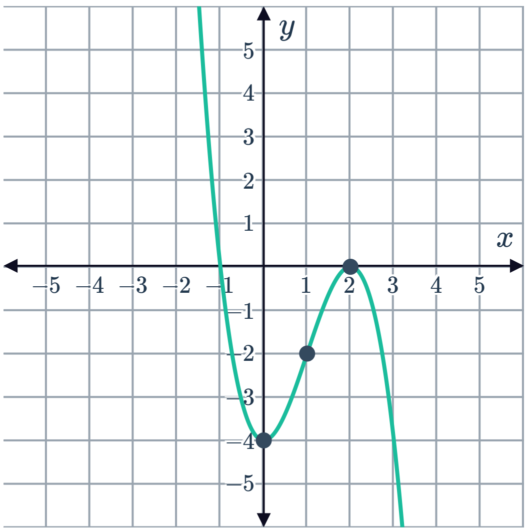
12

a

i $(2,0)$ is a local maximum and $(0,-4)$ is a local minimum.

ii $(1,-2)$

iii



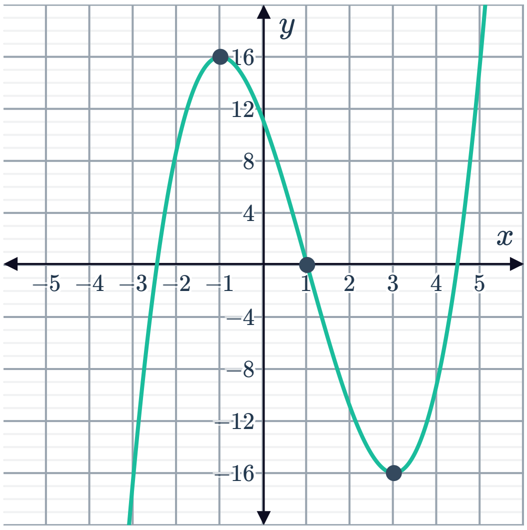
iv Concave up on $(-\infty,1)$ since $f''(x)$ is positive and concave down on $(1,\infty)$ since $f''(x)$ is negative

b

i $(-1, 16)$ is a local maximum and $(3, -16)$ is a local minimum.

ii $(1, 0)$

iii



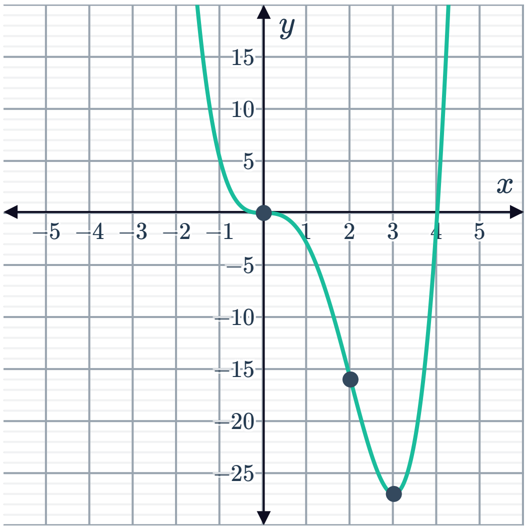
iv Concave up on $(1, \infty)$ since $f''(x)$ is positive and concave down on $(-\infty, 1)$ since $f''(x)$ is negative

c

i $(0, 0)$ is a horizontal point of inflection and $(3, -27)$ is the global minimum.

ii $(0, 0), (2, -16)$

iii

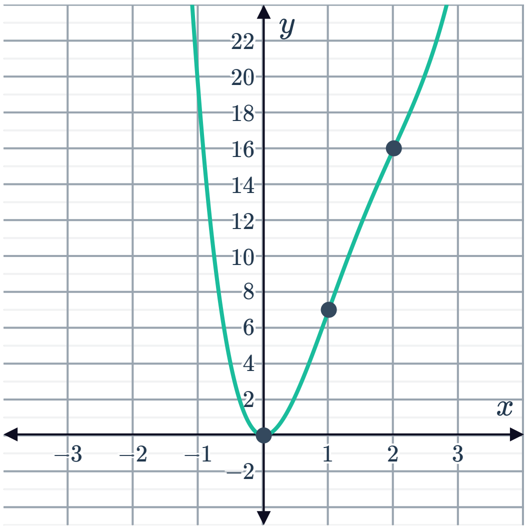


iv Concave up on $(-\infty, 0)$ and $(2, \infty)$ since $f''(x)$ is positive and concave down on $(0, 2)$ since $f''(x)$ is negative d

i $(0,0)$ is the global minimum.

ii $(1,7),(2,16)$

iii



iv Concave up on $(-\infty,1)$ and $(2,\infty)$ since $f''(x)$ is positive and concave down on $(1,2)$ since $f''(x)$ is negative

13

a $(0,253)$

b $\left(-\sqrt[3]{\dfrac{3}{2}}-5,0\right)$

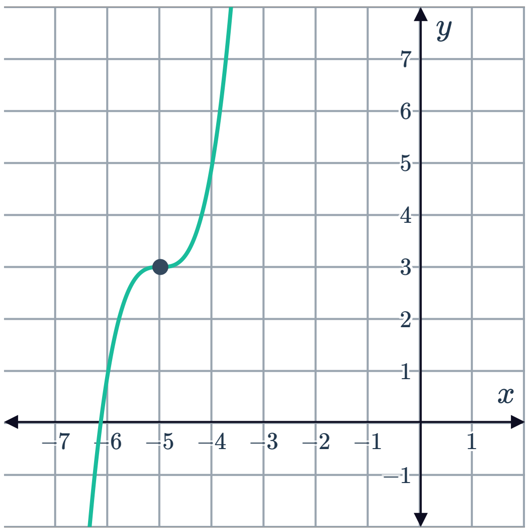
c $(-5,3)$

d $(-5,3)$ is a horizontal point of inflection.

e As $f''(x)=0$ and the concavity changes from negative to positive, there is a point of inflection at $x=-5$. The point of inflection is $(-5,3)$.

f As $x\to\infty, y\to\infty$ and as $x\to-\infty, y\to-\infty$.

g



a $(0,2)$

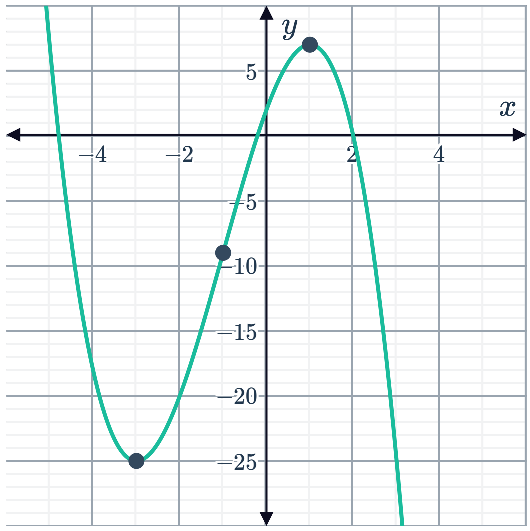
b $(-3,-25),(1,7)$

c $(-3,-25)$ is a local minimum and $(1,7)$ is a local maximum.

d As $f''(x)=0$ and the concavity changes from positive to negative, there is a point of inflection at $x=-1$. The point of inflection is $(-1,-9)$.

e As $x \to \infty, y \to -\infty$ and as $x \to -\infty, y \to \infty$.

f



15

a $(0,-16)$

b $(-2,0),(1,0)$

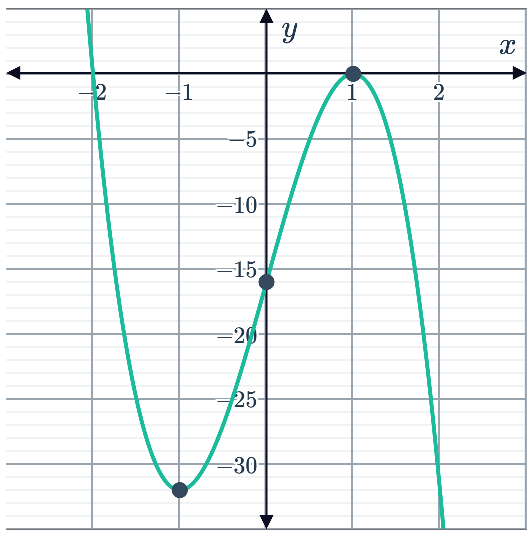
c $(-1,-32),(1,0)$

d $(-1,-32)$ is a local minimum and $(1,0)$ is a local maximum.

e As $f''(x)=0$ and the concavity changes from positive to negative, there is a point of inflection at $x=0$. The point of inflection is $(0,-16)$.

f As $x \to \infty, y \to -\infty$ and as $x \to -\infty, y \to \infty$.

g



16 **a** $(0,16)$

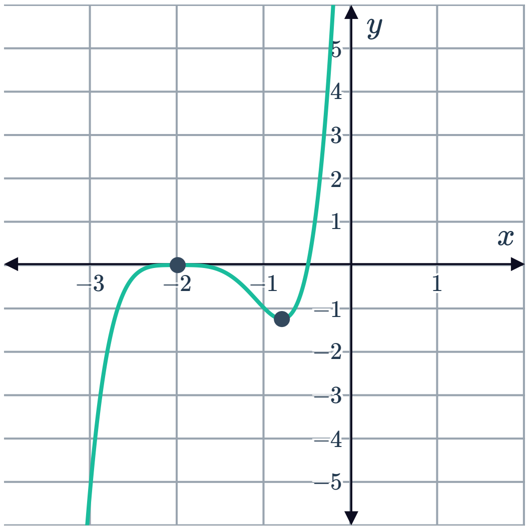
b $(-2,0),\left(-\frac{1}{2},0\right)$

c $(-2,0),\left(-\frac{4}{5},-\frac{3888}{3125}\right)$

d $(-2,0)$ is a local maximum and $\left(-\frac{4}{5},-\frac{3888}{3125}\right)$ is a local minimum.

e As $x \to \infty, y \to \infty$ and as $x \to -\infty, y \to -\infty$.

f



17

a $(0,2)$

b None

c Even

d $(0,2)$

e $(0,2)$ is the global maximum.

f As $x \to \infty, y \to 0$ and as $x \to -\infty, y \to 0$.

g